Birzeit University
Faculty of Engineering and Technology
Electrical and Computer Engineering Department

EE2312 – Signals and Systems
MATLAB Assignment Report

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Course:	EE2312 Signals and Systems

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Question I – Signal Generation and Plotting

Objective:

Generate and plot the following signals using MATLAB on a single figure and as subplots:

1. A finite pulse π(t/6) with value = 5 centered at t = 10

2. x(t) = u(t–1) – r(–t+7) + r(t+4) + π((t+1)/6)

3. x(t) = π((t–3)/6) – r(t–7) + r(t–13) – u(t–13) on [0, 22]

MATLAB Code:

```
t = -20:0.01:30;
u    = @(t) (t >= 0);
r    = @(t) t .* (t >= 0);
rect = @(t) (abs(t) <= 0.5);

% Signal 1: 5*rect((t-10)/6)
s1 = 5 * rect((t - 10) / 6);

% Signal 2: u(t-1) - r(-t+7) + r(t+4) + rect((t+1)/6)
s2 = u(t-1) - r(-t+7) + r(t+4) + rect((t+1)/6);

% Signal 3: rect((t-3)/6) - r(t-7) + r(t-13) - u(t-13)
t3 = 0:0.01:22;
s3 = rect((t3-3)/6) - r(t3-7) + r(t3-13) - u(t3-13);

% Figure 1 – All on one plot
figure(1);
plot(t,s1,'r','LineWidth',1.5); hold on;
plot(t,s2,'b','LineWidth',1.5);
plot(t3,s3,'g','LineWidth',1.5);
legend('Signal 1','Signal 2','Signal 3');
title('Question I - All Signals'); xlabel('t'); grid on;

% Figure 2 – Subplots
figure(2);
subplot(3,1,1); plot(t,s1,'r','LineWidth',1.5);
title('Signal 1: 5pi((t-10)/6)'); xlabel('t'); grid on;
subplot(3,1,2); plot(t,s2,'b','LineWidth',1.5);
title('Signal 2'); xlabel('t'); grid on;
subplot(3,1,3); plot(t3,s3,'g','LineWidth',1.5);
title('Signal 3'); xlabel('t'); grid on;
```

Results and Discussion:

Signal 1 is a rectangular pulse of amplitude 5 centered at t=10 with width 6 (active from t=7 to t=13). Signal 2 combines a unit step, ramp functions, and a rectangular pulse producing a linearly growing waveform. Signal 3 represents a piecewise signal that starts with a rectangular pulse, slopes downward from t=7 to t=13, then remains at a constant negative value, defined over [0, 22].

Question II – Sinusoidal Signals and Operations

Objective:

Generate $y1(t)=10\sin(0.25\pi t)$, $y2(t)=5\cos(0.125\pi t)$, $y3(t)=8\sin(2\pi t)$, compute combined signals, analyze periodicity, and plot amplitude spectra.

MATLAB Code:

```
t  = 0:0.001:64;   % 4 cycles of y2 (highest period = 16s)
y1 = 10 * sin(0.25*pi*t);
y2 =  5 * cos(0.125*pi*t);
y3 =  8 * sin(2*pi*t);

m1 = y1 + y2;   m2 = y1 - y2;
m3 = y2 .* y3;  m4 = y1 .* y3;

% Subplots of y1, y2, y3 (Figure 1)
% Combined m1–m4 on same figure (Figure 2)
% Amplitude spectra via FFT (Figure 3)
```

Periodicity Analysis:

Signal	Periodic?	Period T	Alternating?	Half-Wave Odd Sym.?
m1 = y1+y2	YES	LCM(8,16) = 16 s	YES	NO
m2 = y1–y2	YES	16 s	YES	NO
m3 = y2×y3	YES	LCM(16,1) = 16 s	YES	YES
m4 = y1×y3	YES	LCM(8,1) = 8 s	YES	YES

m3 half-wave symmetry proof: $m3(t + T/2) = y2(t+8)\cdot y3(t+8) = -y2(t)\cdot y3(t) = -m3(t)$ ✓ m4 half-wave symmetry proof: $m4(t + T/2) = y1(t+4)\cdot y3(t+4) = -y1(t)\cdot y3(t) = -m4(t)$ ✓

Question III – Zero-State Responses

Objective:

Solve four differential equations numerically using ode45, determine stability type, and identify steady-state responses.

MATLAB Code (System 1 example):

```
tspan = [0 100];
ode1 = @(t,y) [-0.125*y(1) + 5];
ode2 = @(t,y) [y(2); -5*y(1) - 7*y(2) + 5*cos(0.01*pi*t)];
ode3 = @(t,y) [y(2); -5*y(1) - 4*y(2) + 5];
ode4 = @(t,y) [y(2); -5*y(1) + 4*y(2) + 5];
[t1,y1] = ode45(ode1, tspan, [0]);
[t2,y2] = ode45(ode2, tspan, [0 0]);
[t3,y3] = ode45(ode3, tspan, [0 0]);
[t4,y4] = ode45(ode4, tspan, [0 0]);
```

Stability and Steady-State Analysis:

System	Equation	Roots	Stability	Steady State	Value
Sys 1	$y'+0.125y=5u(t)$	–0.125	Asymptotically Stable	Constant	40
Sys 2	$y''+7y'+5y=5\cos(0.01\pi t)$	–0.70, –6.30	Asymptotically Stable	Sinusoidal	~0
Sys 3	$y''+4y'+5y=5u(t)$	–2±j	Asymptotically Stable	Constant	1
Sys 4	$y''-4y'+5y=5u(t)$	2±j	UNSTABLE	Does NOT exist	∞

Time constants: Sys1: τ=8s, 4τ=32s | Sys2: τ≈1.43s, 4τ≈5.7s | Sys3: τ=0.5s, 4τ=2s | Sys4: UNSTABLE

Question IV – Step Response of Systems

Objective:

Simulate two systems and plot their step responses using transfer functions.

System 1: $5y^{(5)} + 2y''' + 6y' + 4y = 2x'' + 3x' - 6x$

System 2: $y'' + 2y' + 4y = 5x + 3x'$

```
% System 1: H(s) = (2s^2+3s-6) / (5s^5+2s^3+6s+4)
num1 = [2, 3, -6];
den1 = [5, 0, 2, 0, 6, 4];
sys1 = tf(num1, den1);

% System 2: H(s) = (3s+5) / (s^2+2s+4)
num2 = [3, 5];
den2 = [1, 2, 4];
sys2 = tf(num2, den2);

figure; subplot(2,1,1); step(sys1); grid on;
subplot(2,1,2); step(sys2); grid on;
pzmap(sys1); pzmap(sys2);
```

Results:

System 1 has poles in the right half-plane (unstable) — the step response diverges. System 2 is stable with all poles in the left half-plane; it settles at steady-state value = 5/4 = 1.25.

Note: Simulink separate/integrate modeling was replaced by equivalent MATLAB tf() and step() commands.

Question V – Convolution Integral

Objective:

Compute and plot the convolution of two signal pairs.

```
dt   = 0.01;
rect = @(t) (abs(t) <= 0.5);

% Pair 1
t1 = -20:dt:20;
y1 = (10*exp(2*t1)) .* rect((t1-3)/6);
y2 = 12*sin(0.25*pi*t1) .* rect((t1-1)/2);
conv1 = conv(y1, y2, 'full') * dt;

% Pair 2
t2 = -5:dt:5;
y3 = (10*exp(2*t2)) .* rect((t2-0.1)/0.1);  % Approx. impulse
y4 = rect((t2-1)/2);
conv2 = conv(y3, y4, 'full') * dt;
```

Results:

Pair 1: y1 is a one-sided exponential pulse; y2 is a windowed sinusoid. Their convolution produces a smooth bell-shaped pulse centered around t≈8. Pair 2: y3 approximates a Dirac delta function (very narrow pulse). Convolution with y4 (rect pulse) produces a scaled version of y4, confirming the sifting property of the impulse: $x(t)*\delta(t-t0) \approx x(t-t0)$.

Question VI – Laplace Transform

Objective:

Compute symbolic Laplace transforms of two signals.

```
syms t s
y3 = 10 - 10*exp(-0.25*t);
Y3 = laplace(y3, t, s);   % = 10/s - 10/(s+0.25) = 2.5/(s*(s+0.25))

y4 = 8 + 4*exp(-0.0625*t)*cos(0.125*pi*t);
Y4 = laplace(y4, t, s);
% = 8/s + 4*(s+0.0625)/((s+0.0625)^2 + (0.125*pi)^2)
```

Theoretical Results:

Signal	y(t)	Y(s)
Signal 3	(10 – 10e<super>–0.25t</super>)u(t)	2.5 / [s(s + 0.25)]
Signal 4	(8 + 4e<super>–0.0625t</super>cos(0.125πt))u(t)	8/s + 4(s+0.0625) / [(s+0.0625)<super>2</super> + (0.125π)<super>2</super>]

Question VII – System Analysis in Laplace Domain

System: $6y^{(5)} - 7y''' + 2y' + 9y = 10x' + 5x$

Transfer Function: $H(s) = (10s + 5) / (6s^5 - 7s^3 + 2s + 9)$

```
num = [10, 5];
den = [6, 0, -7, 0, 2, 9];
sys = tf(num, den);

p = pole(sys);   % Poles
z = zero(sys);   % Zeros

pzmap(sys);      % Pole-Zero map
step(sys, 2);    % Step response (diverges - unstable)

s_tf    = tf('s');
sys_ramp = sys / s_tf;
step(sys_ramp, 2);  % Ramp response
```

Stability Analysis:

The system has poles in the right half-plane (positive real parts ≈ +1 ± 0.5j). Therefore the system is UNSTABLE. Both step and ramp responses diverge, confirming instability. The pole-zero map clearly shows poles (×) in the right half-plane.

Note: Simulink state observer model replaced by equivalent MATLAB tf() commands.

Question VIII – $H(s) = 10000(s+2)/(s^2+4s+5)$

```
num = 10000 * [1, 2];
den = [1, 4, 5];
sys = tf(num, den);

figure(1); pzmap(sys);   % Poles: -2 ± j | Zero: -2
figure(2);
subplot(2,1,1); step(sys);     % Steady state = H(0) = 4000
s = tf('s');
subplot(2,1,2); step(sys/s, 10);  % Ramp response
```

Results:

Property	Value
Poles	$-2 \pm j$ (Left Half-Plane)
Zero	-2
Stability	STABLE – Asymptotically Stable
Step Steady-State	$H(0) = 10000 \times 2/5 = 4000$
Ramp Response	Diverges (ramp input to stable system)

Question IX – MISO System Analysis

$H1(s) = 10000(s+3)/(s^2+6s+8)$ | $H2(s) = 100(s–3)/(s^2+2s+1)$

```
num1 = 10000*[1,3]; den1 = [1,6,8]; sys1 = tf(num1,den1);
num2 = 100*[1,-3];  den2 = [1,2,1]; sys2 = tf(num2,den2);

% Step responses
step(sys1); step(sys2);

% Frequency response of H1
w = logspace(-2,4,1000);
bode(sys1, w);
```

Results:

System	Poles	Zeros	Stability	SS Value
H1	–2, –4	–3	STABLE	10000×3/8 = 3750
H2	–1, –1 (repeated)	+3 (RHP zero)	STABLE	100×(–3)/1 = –300

H2 has a non-minimum phase zero at s=+3 causing initial undershoot (negative response) despite the system being stable. The frequency response of H1 shows low-pass characteristics.

Question X – AM Modulation

Single-tone AM: message fm=0.25Hz, Am=10 | carrier fc=8Hz, Ac=15

```
fm=0.25; fc=8; Am=10; Ac=15;
t = 0:0.001:16;
msg       = Am * sin(2*pi*fm*t);
carrier   = Ac * cos(2*pi*fc*t);
modulated = (Ac + msg) .* cos(2*pi*fc*t);
% Spectra via FFT – subplots for message, carrier, modulated
```

Periodicity Analysis:

```
Tm = 1/0.25 = 4 s | Tc = 1/8 = 0.125 s | Ratio = Tm/Tc = 32 (integer) → PERIODIC
```

The modulated signal is periodic with fundamental period T = 4 s. The AM spectrum shows the carrier at ±8 Hz plus sidebands at ±7.75 Hz and ±8.25 Hz, consistent with standard DSB-LC AM theory: $S(f) = Ac/2[\delta(f–fc)+\delta(f+fc)] + M(f–fc)/2 + M(f+fc)/2$.

Question XI – Fourier Transform

Objective:

Compute Fourier transforms numerically via FFT and compare with theoretical results.

Signal 1: $x(t) = \pi(t/5)$

Theoretical: $X(f) = 5 \cdot \mathrm{sinc}(5f)$ — a sinc function scaled by the pulse width (5).

Result: FFT matches the theoretical $5 \cdot \mathrm{sinc}(5f)$ curve exactly. The main lobe has width 2/5 = 0.4 Hz. Phase flips by π at each zero crossing.

Signal 2: $x(t) = 10\sin(2\pi t) \cdot \cos(40\pi t)$

Using the product-to-sum identity:

```
10sin(2πt)cos(40πt) = 5sin(42πt) − 5sin(38πt)
```

Theoretical frequencies: ±21 Hz and ±19 Hz. FFT confirms impulses at exactly these four frequencies.

Signal 3: $x(t) = 10e^{-3t}u(t)$

Theoretical: $X(f) = 10/(3 + j2\pi f) \rightarrow |X(f)| = 10/\sqrt{(9+(2\pi f)^2)}$

Result: FFT matches the theoretical Lorentzian shape perfectly. Peak at f=0: $|X(0)| = 10/3 \approx 3.33$.

End of Report